# AN EXPLICIT COMPLEX-LEVEL MAP REALISES THE ADJUNCTION BETWEEN DERIVED BASE CHANGE AND RESTRICTION

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

*Editorial note.* The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

*Editorial note.* The author has a trailing space after the sentence ending 'representing $M$.' in the counted proof span; that is the author's text and is delivered as it stands. The single reference inside the counted proof span is recorded as a non-load-bearing navigation pointer.

*Allowed standard prerequisites (editorial).*

- K-injective and K-flat complexes and their use as representatives in the derived category. used at proof lines 28772-28775. Supplies the explicit complexes on which the comparison map is written down.
- The tensor-Hom adjunction: derived base change is left adjoint to restriction. invoked at proof lines 28792-28795. Identifies the cohomology of the comparison map with the adjunction comparison.
- Precomposition and postcomposition on Hom complexes. used at proof lines 28781-28787. Is the degreewise formula for the map.

*Source reference.* Original lemma 14754: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14754–14754. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

**Lemma 0.2.** *Let $R \to R'$ be a ring map. For $K \in D(R)$ and $M \in D(R')$ there is a canonical isomorphism*

$$R\operatorname{Hom}_R(K, M) = R\operatorname{Hom}_{R'}(K \otimes^{\mathbf{L}}_R R', M)$$

*Proof.* Choose a K-injective complex of $R'$-modules $J^\bullet$ representing $M$. Choose a quasi-isomorphism $J^\bullet \to I^\bullet$ where $I^\bullet$ is a K-injective complex of $R$-modules. Choose a K-flat complex $K^\bullet$ of $R$-modules representing $K$. Consider the map

$$\operatorname{Hom}^\bullet(K^\bullet \otimes_R R', J^\bullet) \longrightarrow \operatorname{Hom}^\bullet(K^\bullet, I^\bullet)$$

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The map on degree $n$ terms is given by the map

$$\prod_{n=p+q} \mathrm{Hom}_{R'}(K^{-q} \otimes_R R', J^p) \longrightarrow \prod_{n=p+q} \mathrm{Hom}_R(K^{-q}, I^p)$$

coming from precomposing by $K^{-q} \to K^{-q} \otimes_R R'$ and postcomposing by $J^p \to I^p$. To finish the proof it suffices to show that we get isomorphisms on cohomology groups:

$$\mathrm{Hom}_{D(R)}(K, M) = \mathrm{Hom}_{D(R')}(K \otimes^{\mathbf{L}}_R R', M)$$

which is true because base change $- \otimes^{\mathbf{L}}_R R' : D(R) \to D(R')$ is left adjoint to the restriction functor $D(R') \to D(R)$ by Lemma 14754. $\square$